

Introduction to Biometric Technologies and Applications (II)

Kevin Hernández Díaz
ITE

Email: kevin.hernandez-diaz@hh.se

Course: Biometric Recognition
DI4025, ht2025

Outline

Automatic biometric systems

- Structure
- Working modes (1:1, 1:N, 1:N+1)
- Performance analysis & system errors

Outline

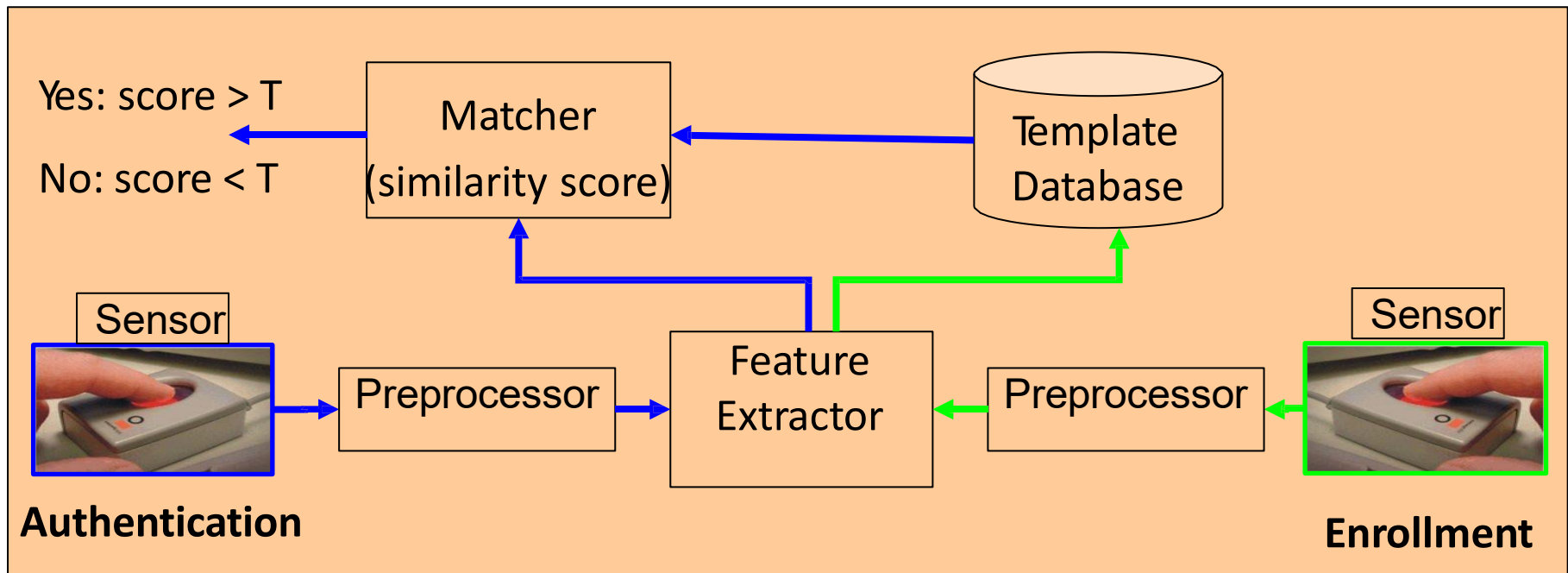
Automatic biometric systems

- Structure
- Working modes (1:1, 1:N, 1:N+1)
- Performance analysis & system errors

Biometric System: Structure

A pattern recognition system that operates by

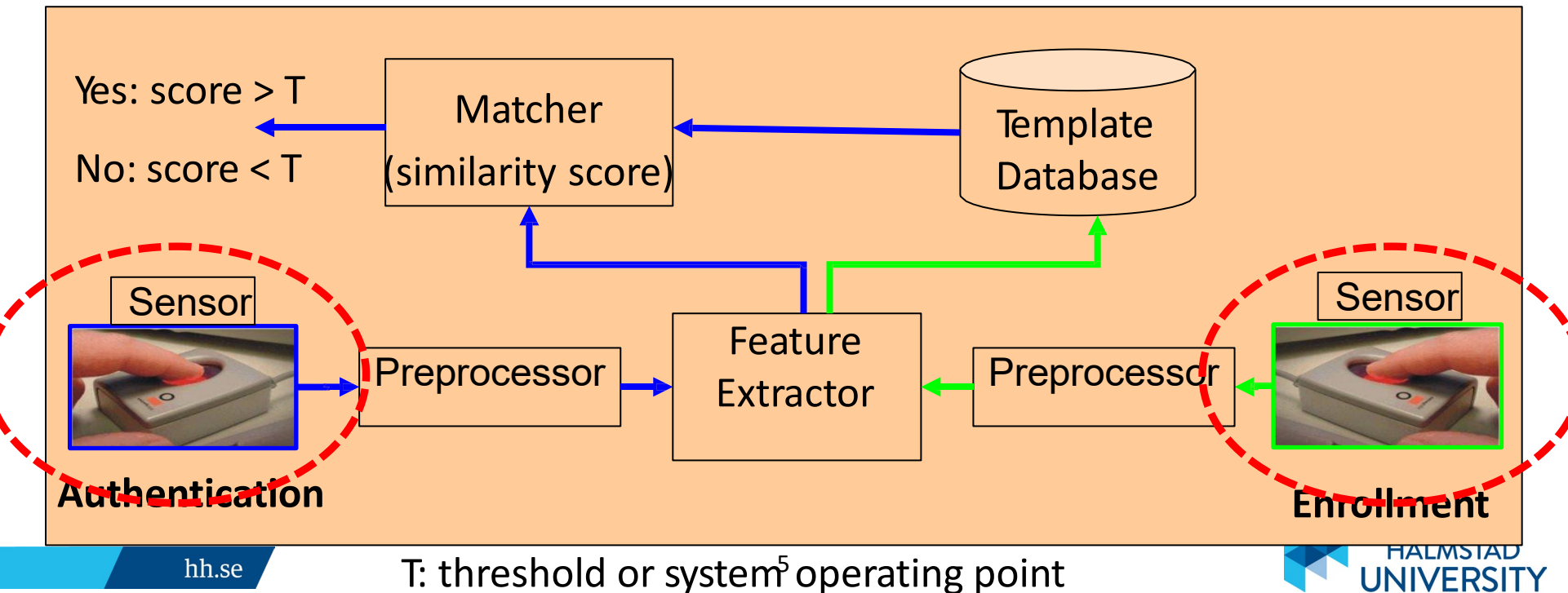
- **Acquiring** biometric data from an individual
- **Extracting** a feature set (characteristics) from the data
- **Comparing** the feature set against template sets in the database
- **Deciding** about the identity of the individual



Biometric System: Structure

Acquisition

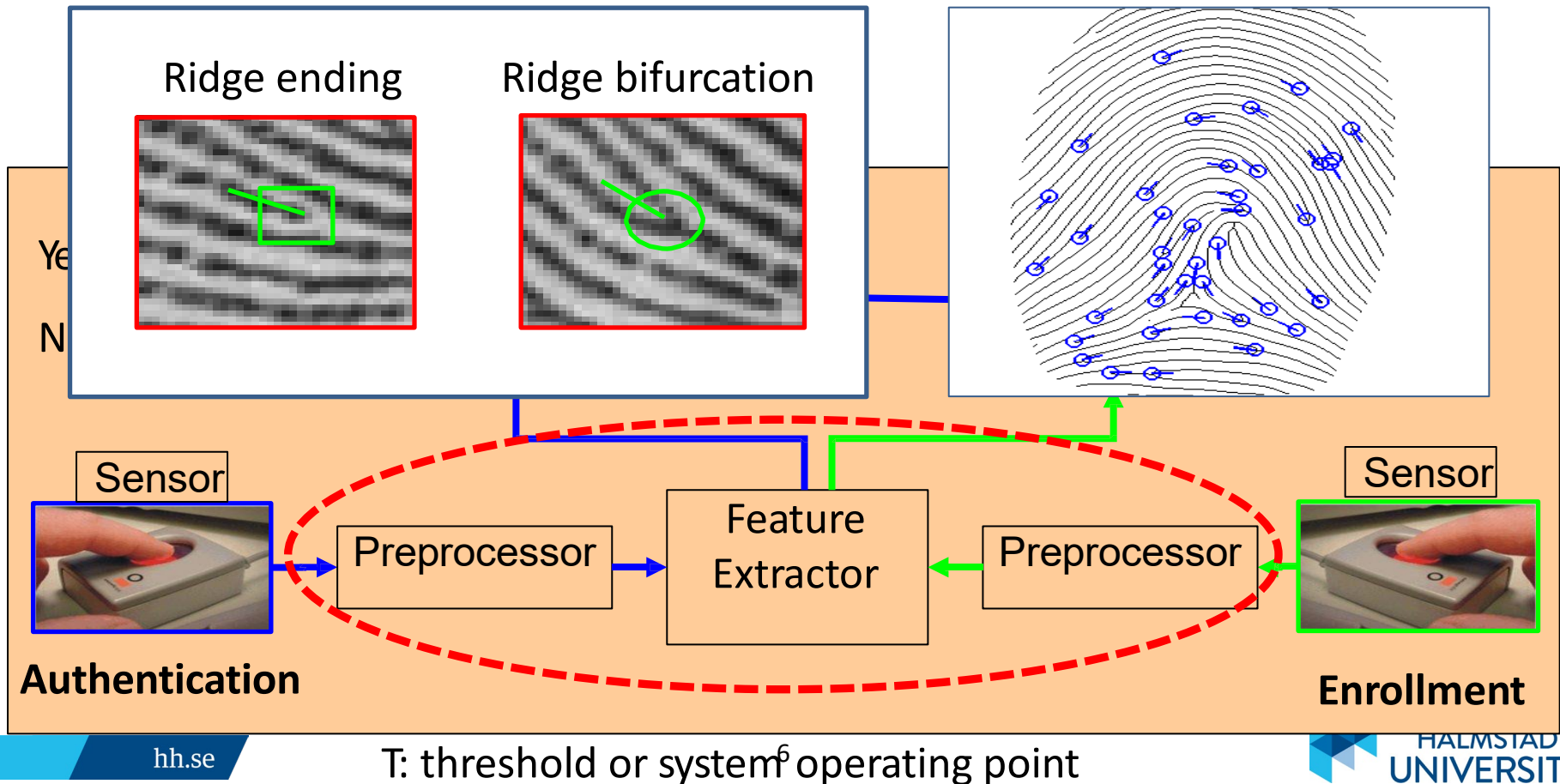
- Sensor that captures the biometric data of an individual and converts it to a **digital signal**
- The capture process is crucial to ensure good **quality** and **quantity** of the acquired data



Biometric System: Structure

Preprocessing and feature extraction

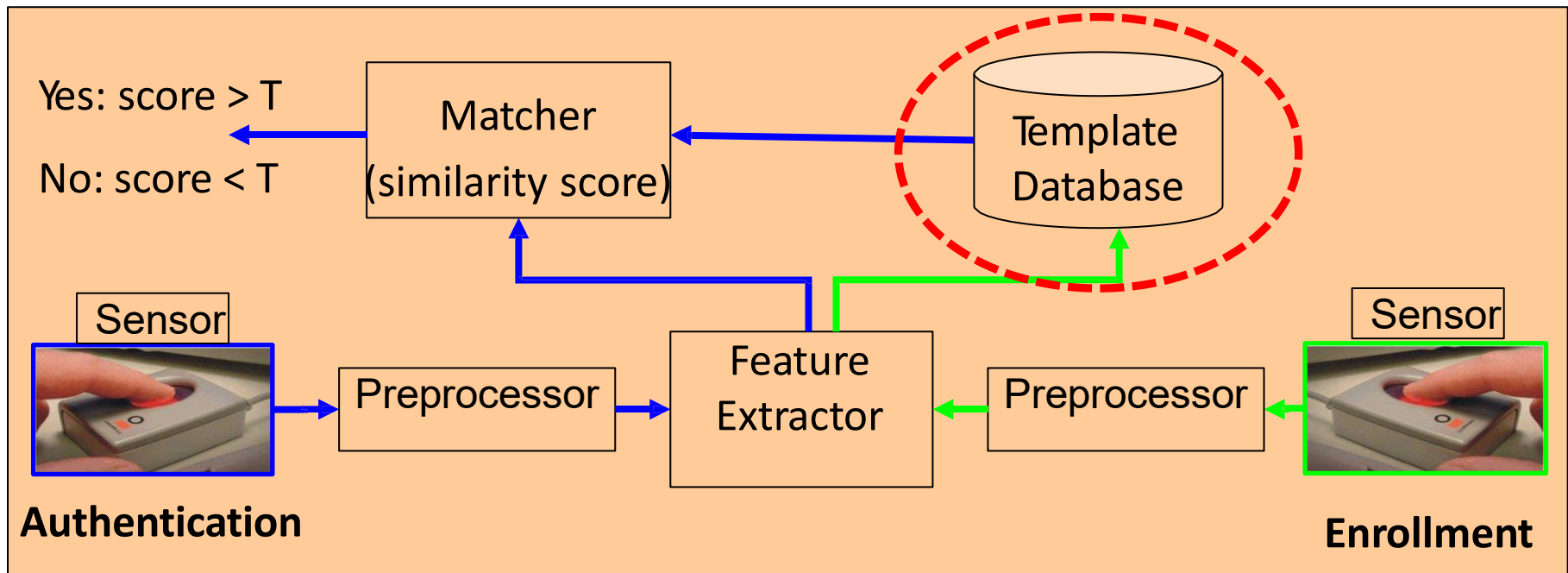
- Noise elimination and extraction of **salient or discriminatory features** that will be used to model identity
- Example: characteristic points (minutiae) in fingerprints



Biometric System: Structure

Database

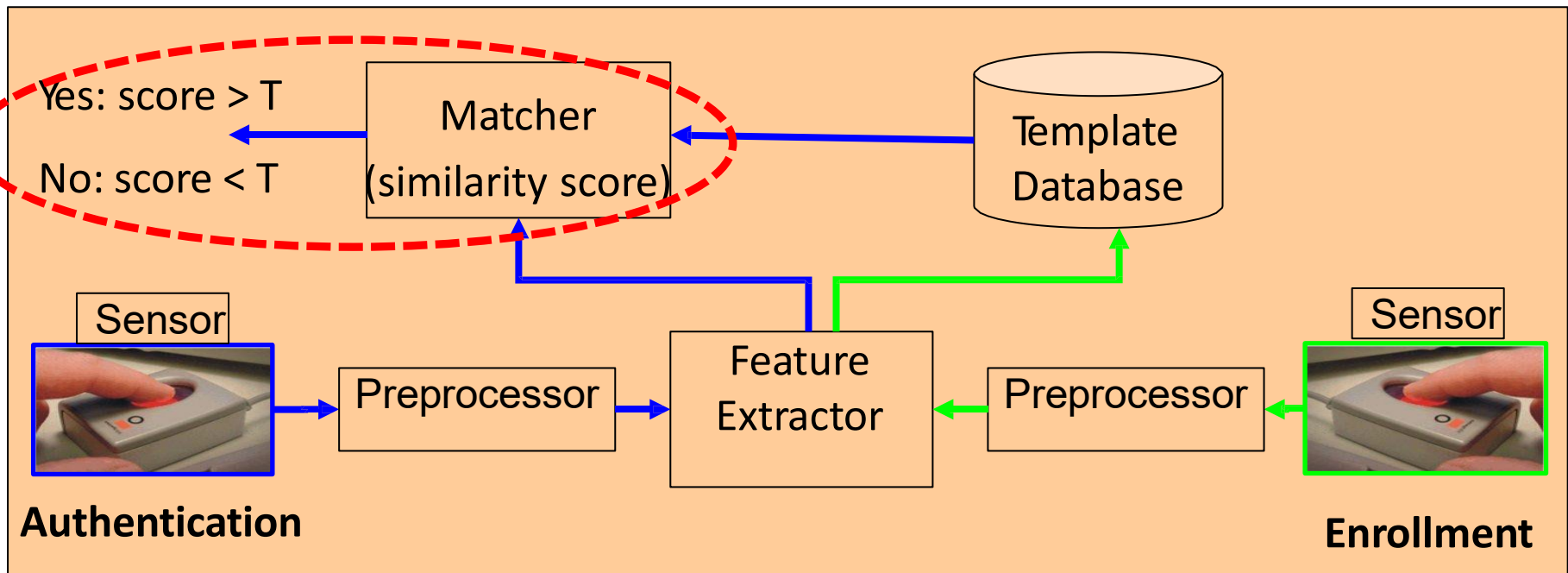
- To store biometric **templates of enrolled users** (authorized)
- **Supervised** or **unsupervised** enrolment
- **Periodic update** and/or capture of **multiple templates** to account for variations observed in the biometric trait
- **Centralized** (server) or **local** (smart card, device)



Biometric System: Structure

Comparison and decision

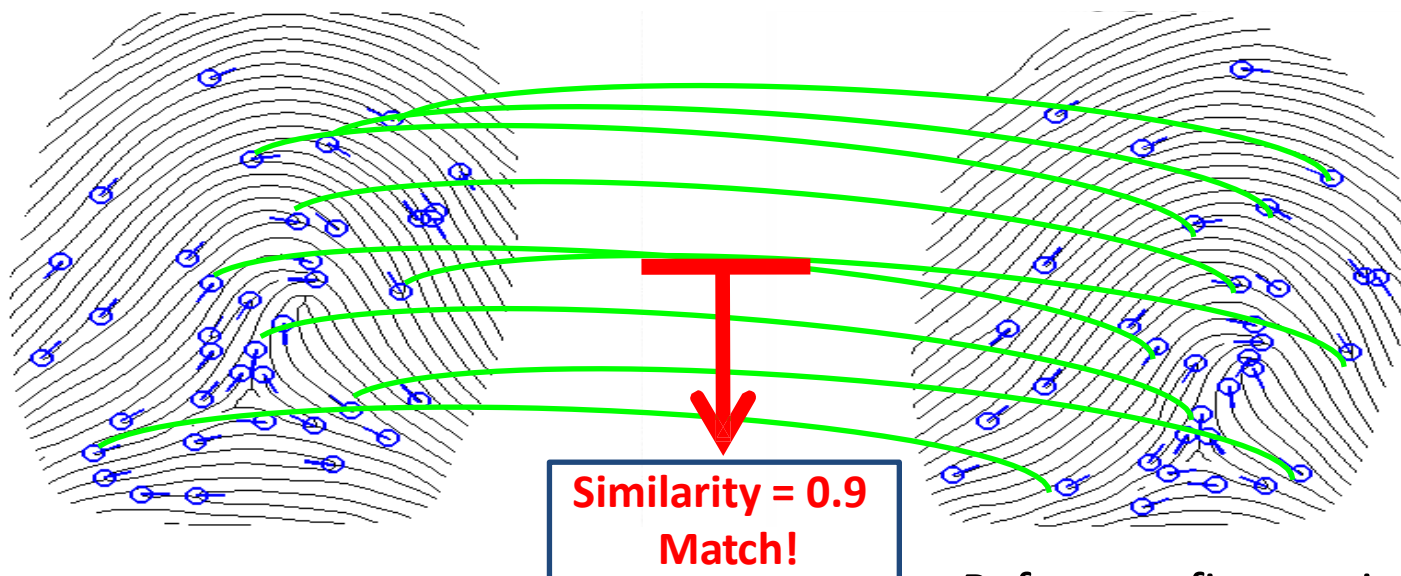
- Compare the newly captured template with templates of enrolled users stored in the database
- Results in **similitude scores**, which can be compared against a **threshold** to decide about the identity of the user



Biometric System: Structure

Comparison and decision

- The output of a biometric matcher is the match score (typically a number), that quantifies the **similarity** between two samples
- The **higher** the score (similarity score), the **more certain** is the system, that the two samples come from the same person
- Example: compare position of characteristic points between two fingerprints



Who made this impression?

Reference fingerprint

Outline

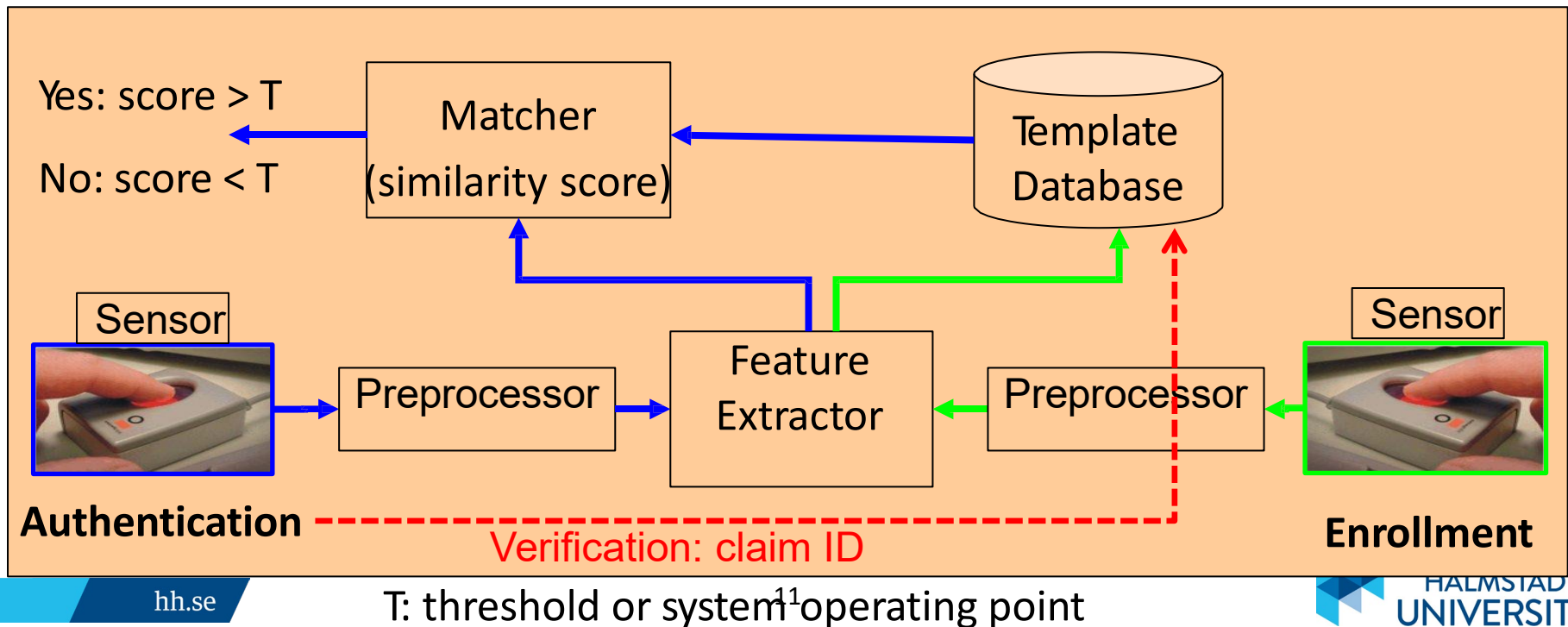
Automatic biometric systems

- Structure
- Working modes (1:1, 1:N, 1:N+1)
- Performance analysis & system errors

Biometric System: Working Modes

Two working modes depending on the application context

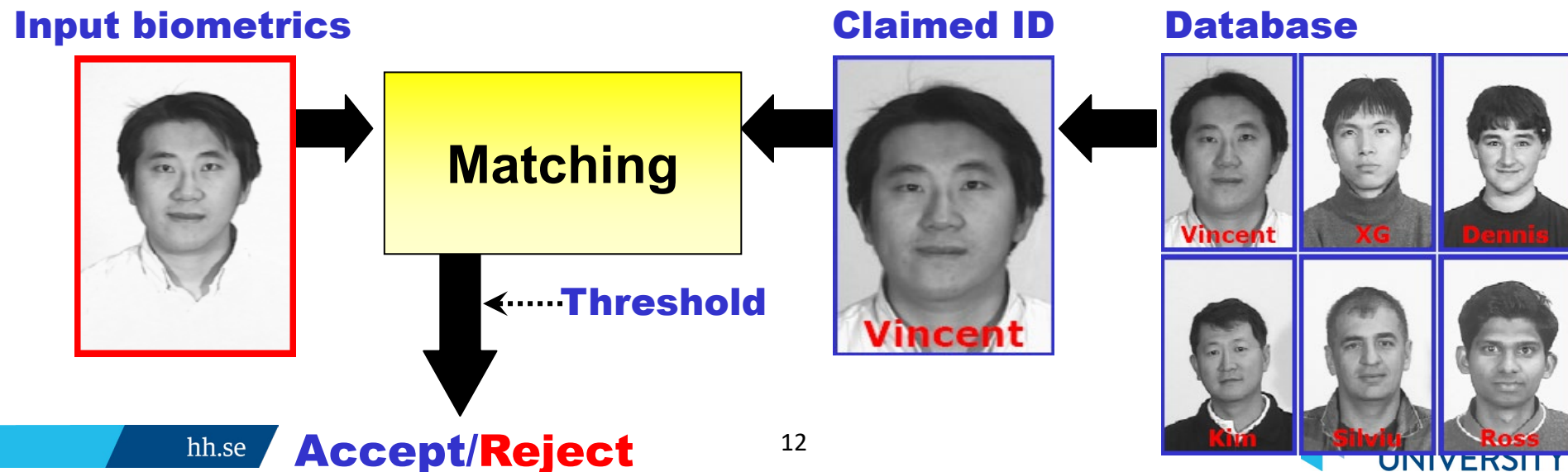
- **Verification:** 1:1 matching (you claim who you are)
- **Identification:** 1:N, 1:N+1 matching (do not claim who you are)



Biometric System: Working Modes

Verification

- **Question:** are you who you say you are? (1:1 matching)
- The person **claims** to be 'Vincent' (via PIN, username or smart card), so the system matches and compare his/her biometrics only with Vincent's stored template.
- If they match, then user is 'verified' that s/he is indeed 'Vincent'
- **Decision:** Score > threshold, then **accept**, otherwise **reject**
- Example: access control application scenarios ("friendly systems")

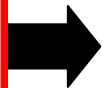


Biometric System: Working Modes

Identification (1:N or closed set)

- **Question: who is this person? (1:N matching)**
- The person does not make any claim about identity
- The system matches the person's biometrics against the **whole database** to figure out the identity by finding the **closest match** or a list of the **N most similar identities**.

Input biometrics



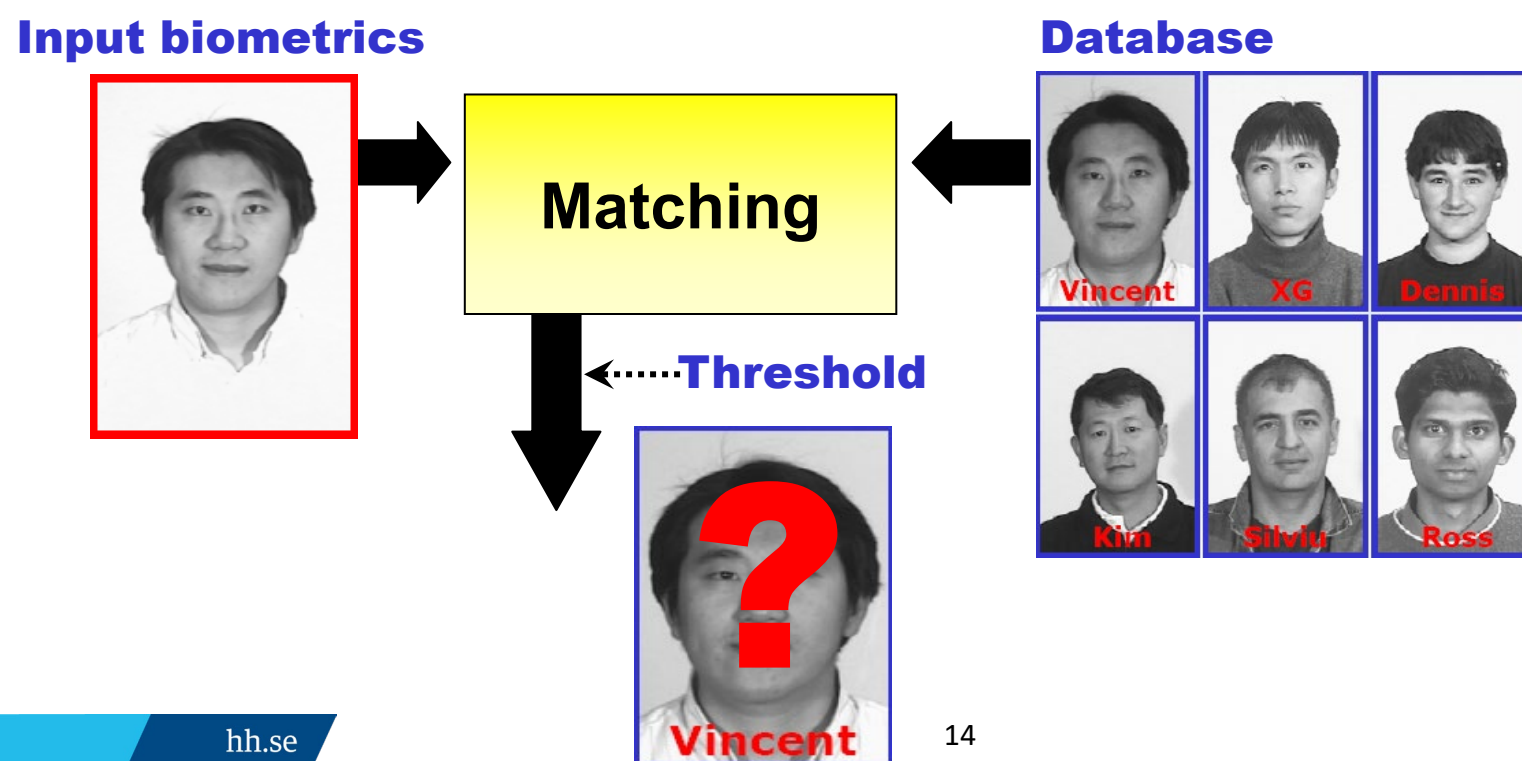
Database



Biometric System: Working Modes

Identification (1:N+1 or open set)

- **Question: is this person in my database? (1:N+1 matching)**
- The closest match is compared against a **threshold**
- If the closest match is below the threshold, the system concludes that the input sample **does not match with any of the users** that are in the database



Biometric System: Working Modes

The following distinction is usually done in the biometrics literature (although sometimes they are inadvertently mixed up)

Generic terms to refer to establishing the identity of a person:

- Authentication
- Recognition

Specific terms to refer to the working modes of a biometric system:

- Verification
- Identification

Outline

Automatic biometric systems

- Structure
- Working modes (1:1, 1:N, 1:N+1)
- Performance analysis & system errors

Performance of Biometric Systems

Biometric systems **are not error free**. Two acquisitions of the same biometric trait are **never equal** (even if captured one after the other)

- **Sensor limitations**, e.g. low resolution/size...
- **Changes** in the biometric characteristics over time (aging)
- **Variations in user interaction/behaviour**, e.g. expression or pose, uncooperative subjects, make-up/accesories...
- **Variations in the environment**, e.g. light, weather...
- **Back-end** processing (compression, algorithm, speed...)

Output of the biometric system is given by a **similarity measure** "s" (score) between two samples:

- Higher score -> higher similarity
- Decision regulated by a **threshold** "T"

Performance of Biometric Systems

Aim:

- Evaluate the reliability of our system in the form of **numerical values or plots**

So we can:

- Assess **new** developments or improvements
- **Compare** different systems
- Determine if the system fulfils our **requirements** and works properly under the intended **conditions** of our application

Performance: Verification Mode

Verification (1:1 matching)

Two types of attempts:

Genuine access: the person claims who s/he really is

- Comparing two samples from the same individual -> **genuine** score

Impostor access: the person claims someone else's identity

- Comparing two samples from different individuals -> **impostor** score

Four possible outputs, two types of errors:

		Output of the system	
		Genuine	Impostor
Claim (attempt) of the user	Genuine	OK	FR=False Rejection FNM=False Non Match
	Impostor	FA=False Acceptance FM=False Match	OK

Performance: Verification Mode

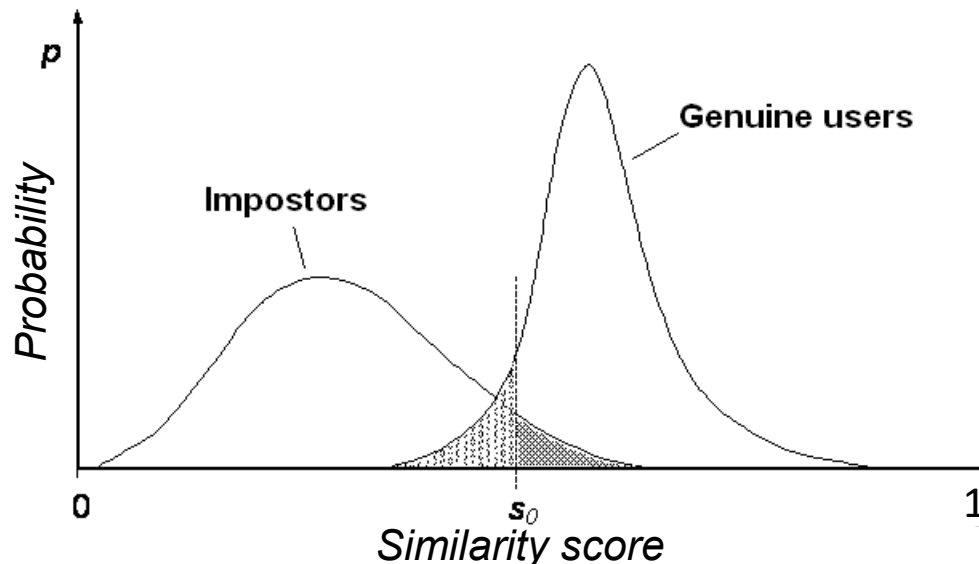
Step 1: compute histogram/distribution of scores

Two distributions of scores "s"

- **Genuine** distribution: generated by genuine attempts
- **Impostor** distribution: generated by impostor attempts

obtained by making many comparisons of the two types of attempts and plotting the **histograms of scores** "s" obtained for the two cases

To do this, we need a **training database of "controlled" data** (i.e. that we know the answer accepted/rejected for each matching)



Ideally the distributions should not overlap, but...

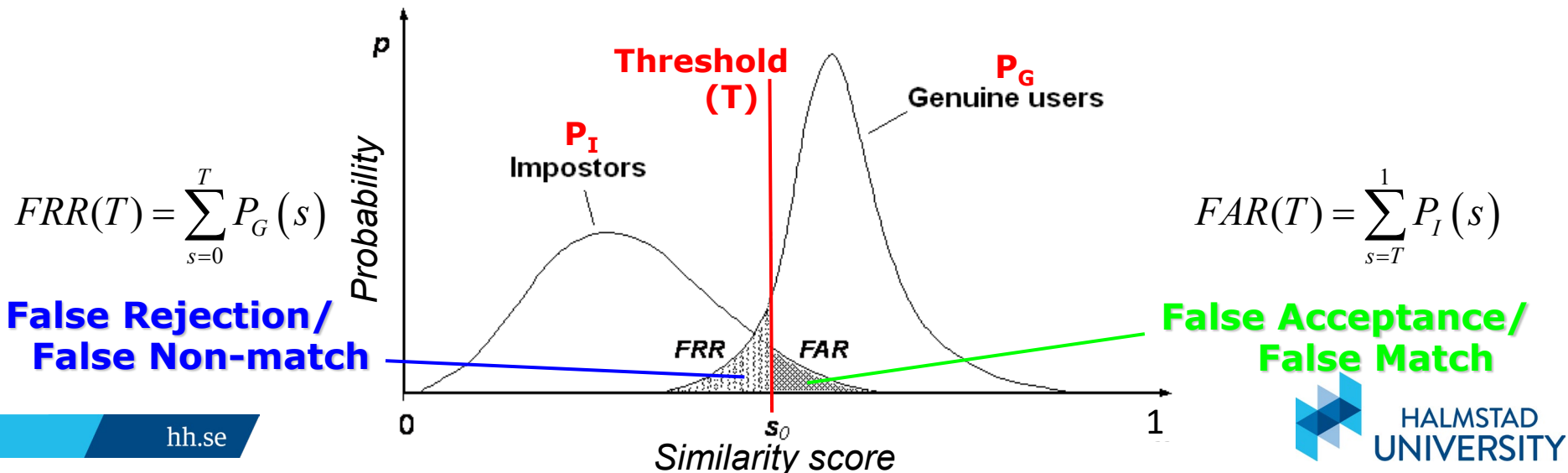
Performance: Verification Mode

Step 2: compute **FAR** and **FRR** at multiple thresholds

Errors for a given threshold T :

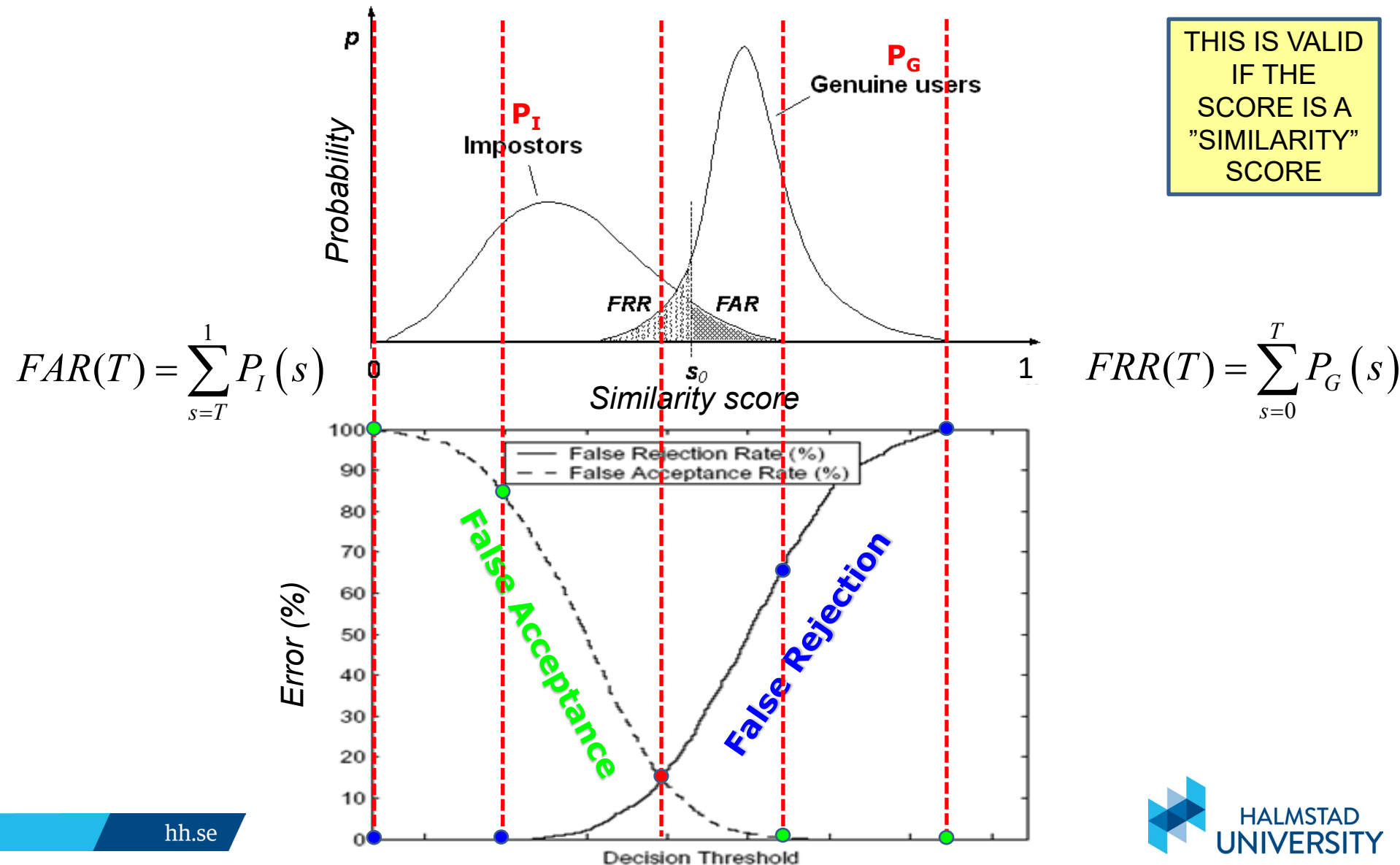
- **FRR**=area of the "genuine" curve below the threshold
 - > proportion of **genuine attempts** being "rejected"
 - > probability that a **genuine user** will be incorrectly rejected
- **FAR**=area of the "impostor" curve above the threshold
 - > proportion of **impostor attempts** being "accepted"
 - > probability that an **impostor** will be incorrectly accepted

THIS IS VALID
IF THE
SCORE IS A
"SIMILARITY"
SCORE



Performance: Verification Mode

Step 2: compute **FAR** and **FRR** at multiple thresholds



Performance: Verification Mode

Step 3: compute ROC/DET curve (plot **FAR** vs. **FRR**)

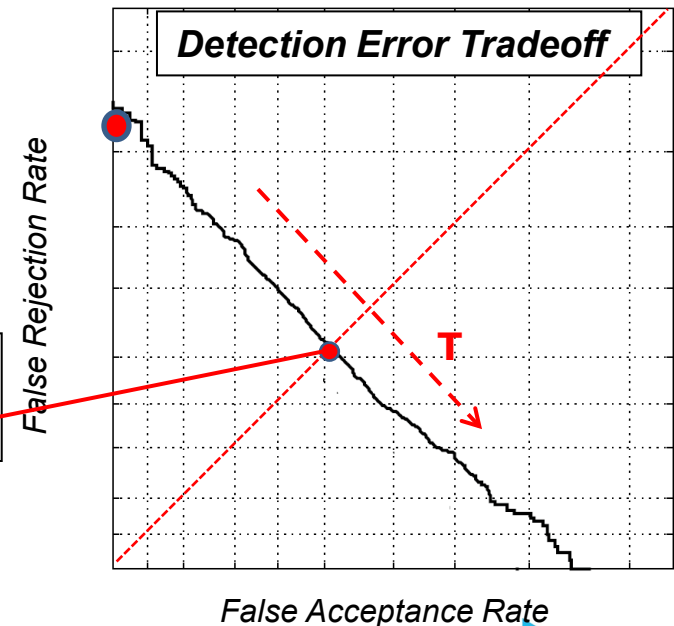
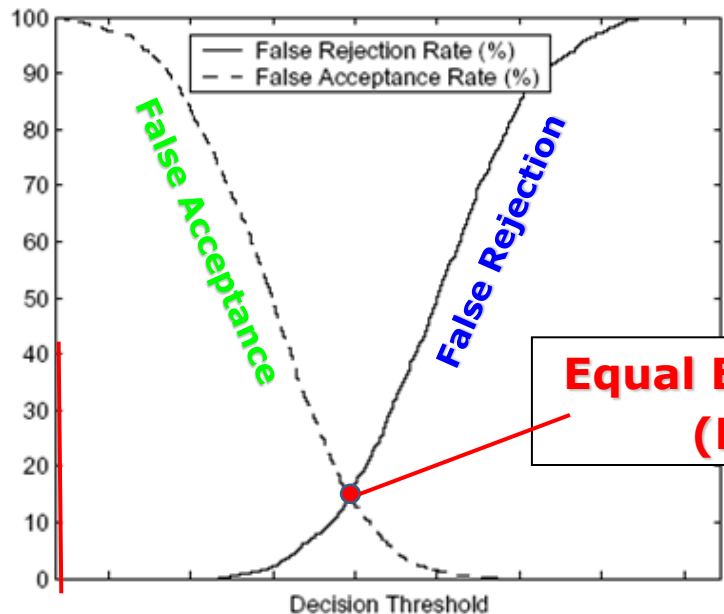
ROC: Receiver Operating Characteristic

DET: Detection Error Tradeoff (ROC curve in logarithmic scale)

THIS IS VALID
IF THE
SCORE IS A
"SIMILARITY"
SCORE

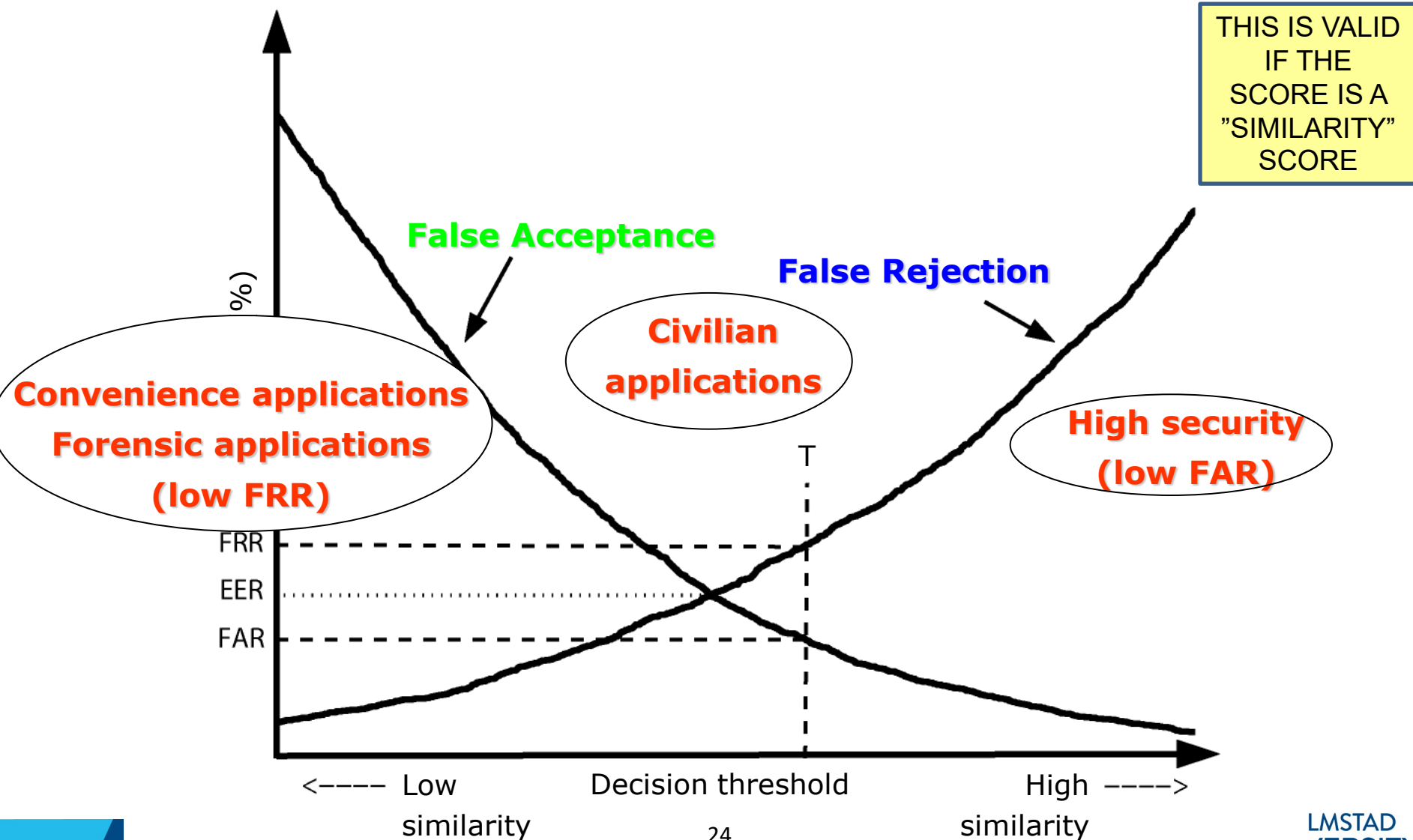
When **FAR** decreases, **FRR** increases and vice-versa

EER is where **FAR**=**FRR**, and measures the degree of overlap between the **FAR** and **FRR** curves



Performance: Verification Mode

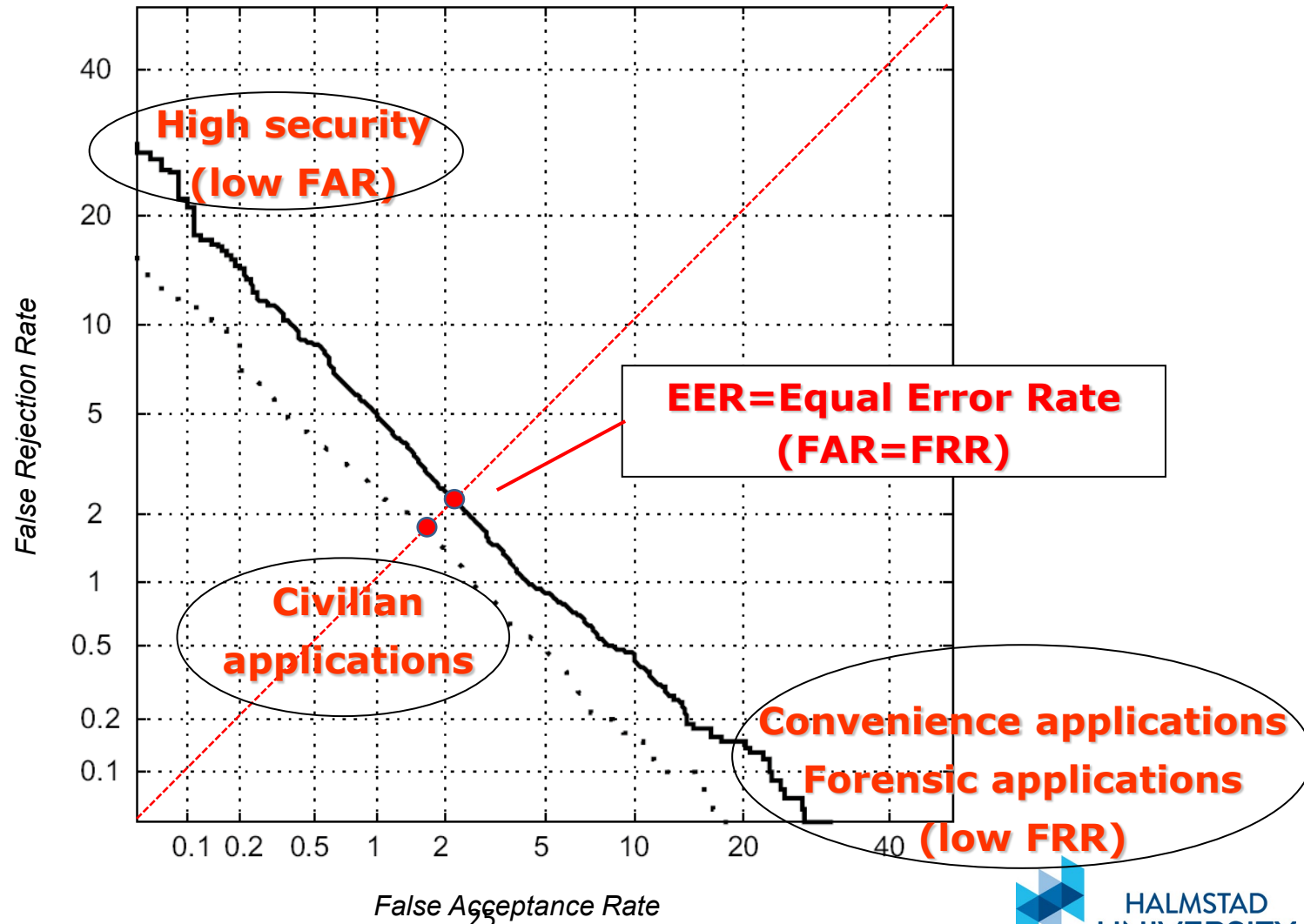
Choosing an appropriate threshold: depends on the **application**!



Performance: Verification Mode

Comparing different systems

Detection Error Tradeoff



Performance: Verification Mode

Summary (things to remember):

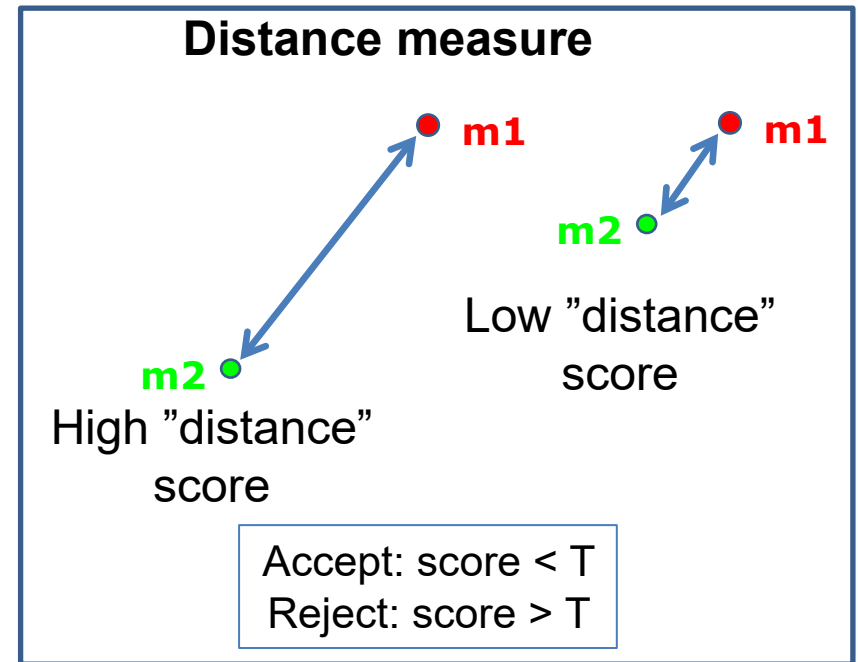
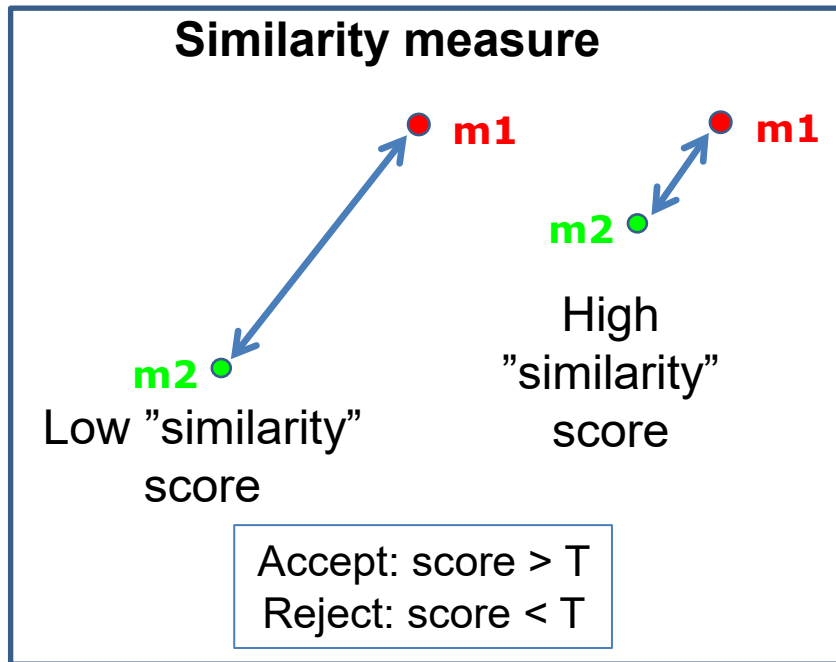
- FAR/FRR affect to **different type of users**
 - FAR: an unauthorized person gets access (security breach)
 - FRR: an authorized person is denied access (inconvenient)
- There is always a **trade-off between FAR/FRR** (opposite behaviour, both cannot be low at the same time)
 - Higher security $\leftrightarrow$ low FAR (increase T, if similarity score)
 - More convenience $\leftrightarrow$ prioritize FRR (decrease T, if similarity score)
- EER (where FAR=FRR) can be used as a single measure to report verification performance, and to **compare** different systems
- With the DET curve, we lose information about the threshold at which each pair of FAR/FRR occurs

Performance: Verification Mode

Similarity vs. distance measures

Instead of a similarity measure, some systems output a distance measure:

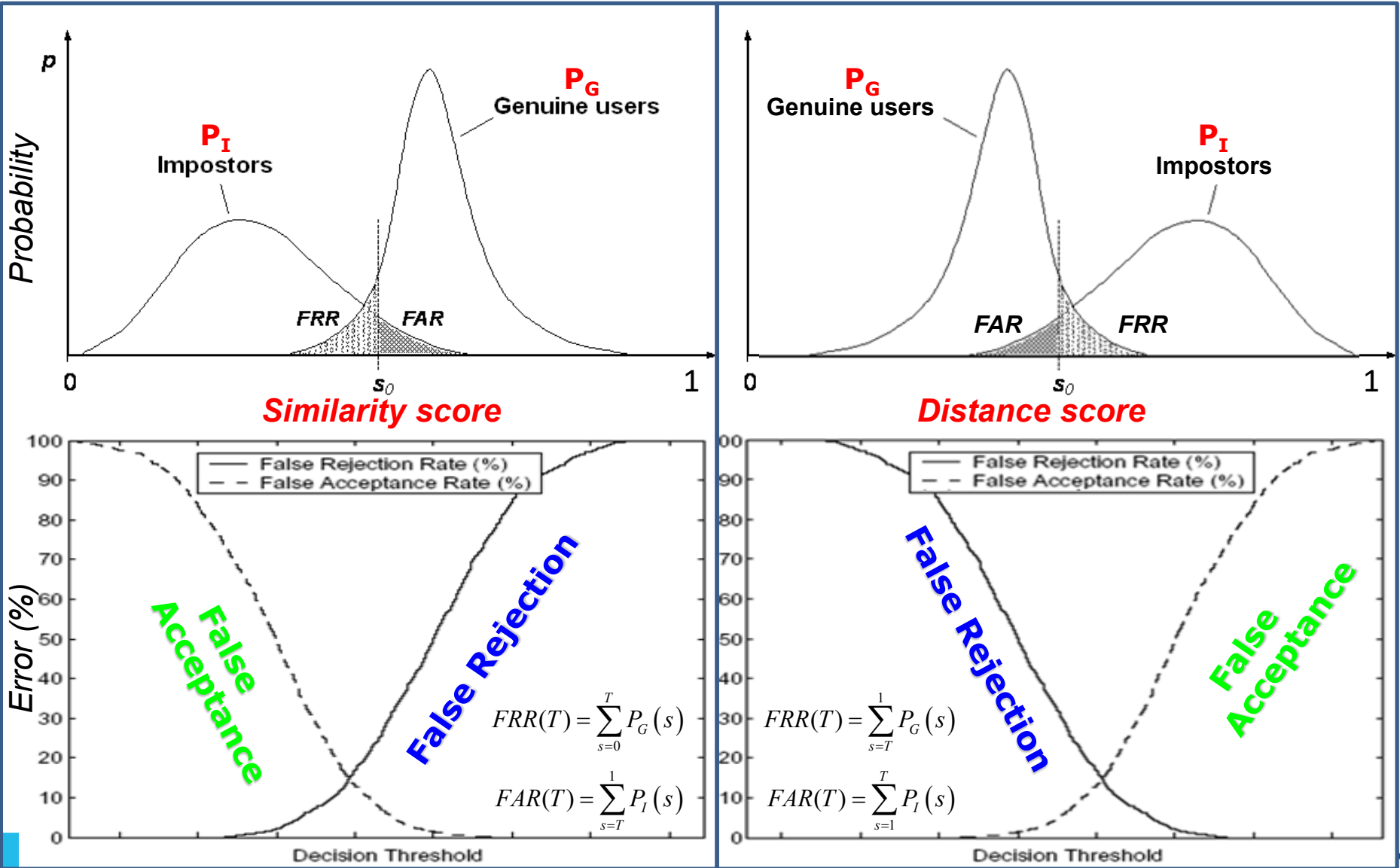
- "Similarity measure": higher score $\rightarrow$ higher similarity
- "Distance measure": lower score $\rightarrow$ higher similarity



In this case, the genuine/impostor curves appear switched

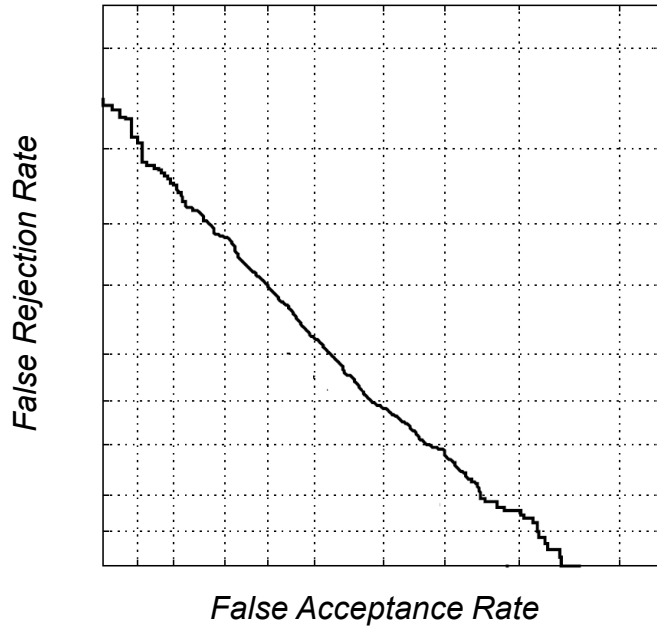
Performance: Verification Mode

Similarity vs. distance measures

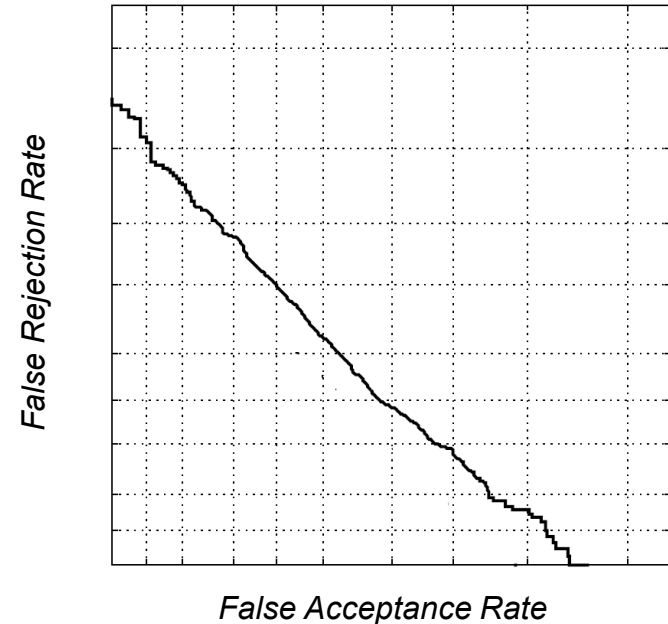


Performance: Verification Mode

Similarity vs. distance measures



Similarity score



Distance score

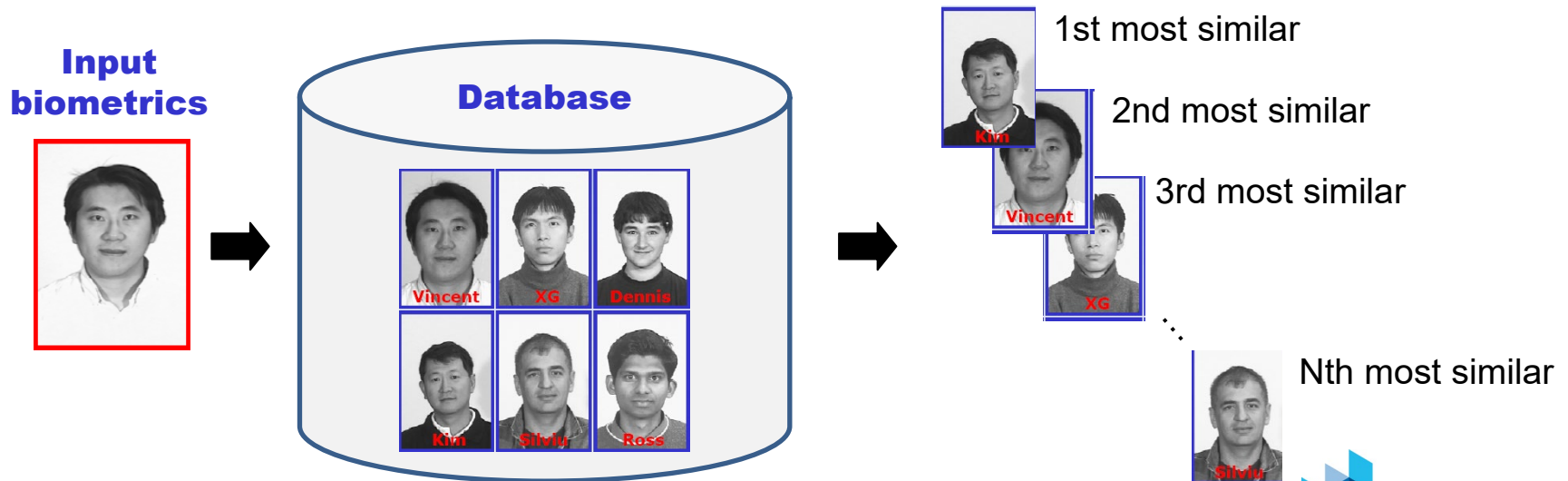
But DETs are the same!

Performance: Identification Mode (1:N)

Identification (1:N matching or closed set)

- The person provides his/her biometric sample, which is compared **against all** stored templates (N) in the system database
- The resulting scores are then sorted and **ranked by the most similar** (i.e. decreasing order of similarity*). The output list is as long as the number of persons in the database

(*) If we use distance measures, then scores are sorted by increasing distance instead, so the first identity has the lowest distance, etc.

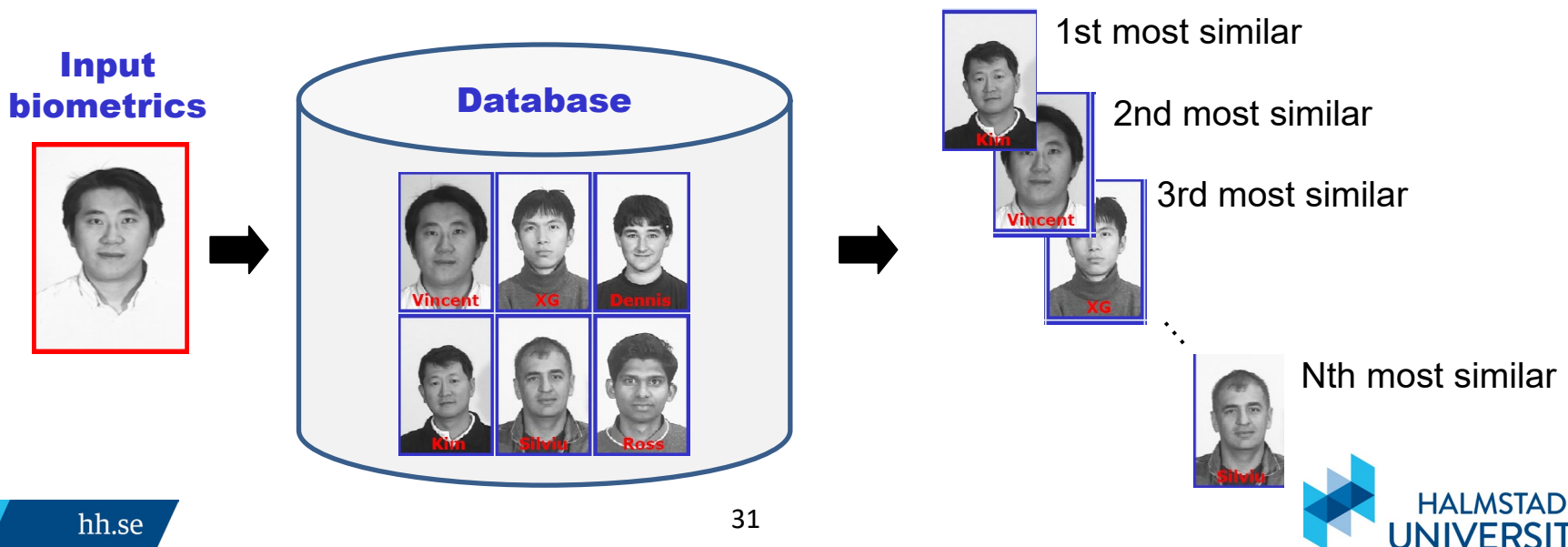


Performance: Identification Mode (1:N)

Identification (1:N matching or closed set)

- **Error**: correct identity not in 1st position (or in the first M positions)
- **Performance measurement**:
 - **True Positive Identification Rate (TPIR)**: probability that the correct identity is within the **top M candidates**
 - **CMC curve**: plot of TPIR against ranks

obtained by making many comparisons, and computing the % of times that the correct identity is in the top M candidates



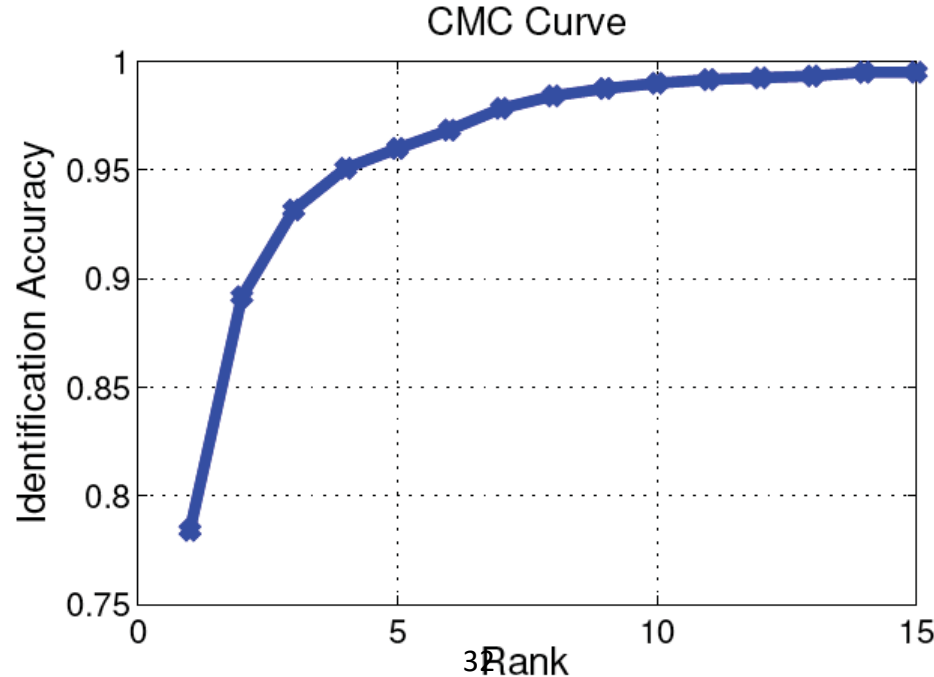
Performance: Identification Mode (1:N)

Identification (1:N matching or closed set)

➤ **CMC curve**: plot of TPIR against ranks

obtained by making many comparisons, and computing the % of times that the correct identity is in the top M candidates

To do this, we need a **training database of "controlled" data** (i.e. that we know the correct identity for each attempt)



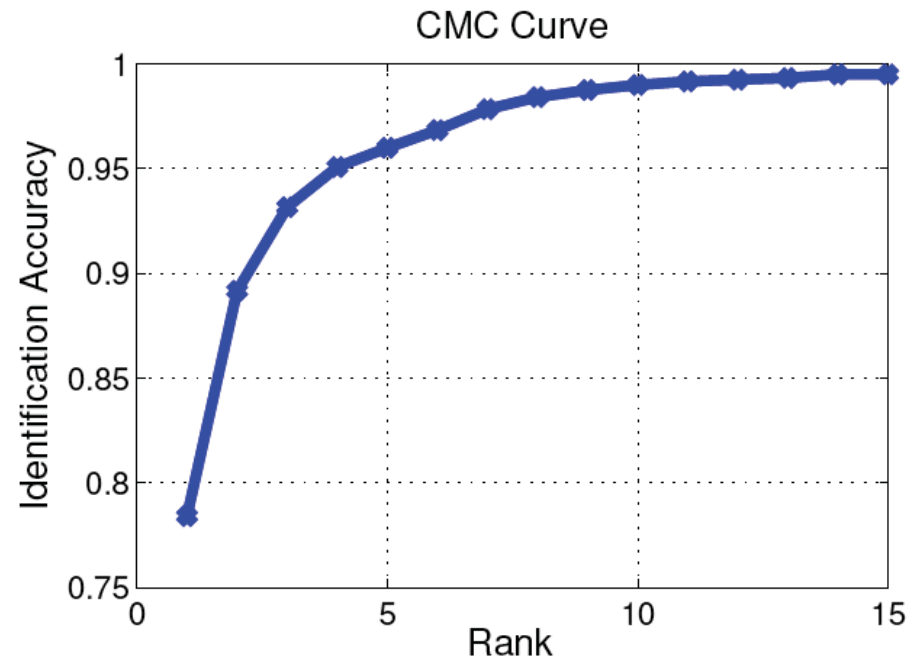
Performance: Identification Mode (1:N)

Identification (1:N matching or closed set)

- In forensic applications: the first M candidates are usually provided, which are then **reviewed manually** by an expert
- Higher M $\rightarrow$ higher probability... but more candidates to examine

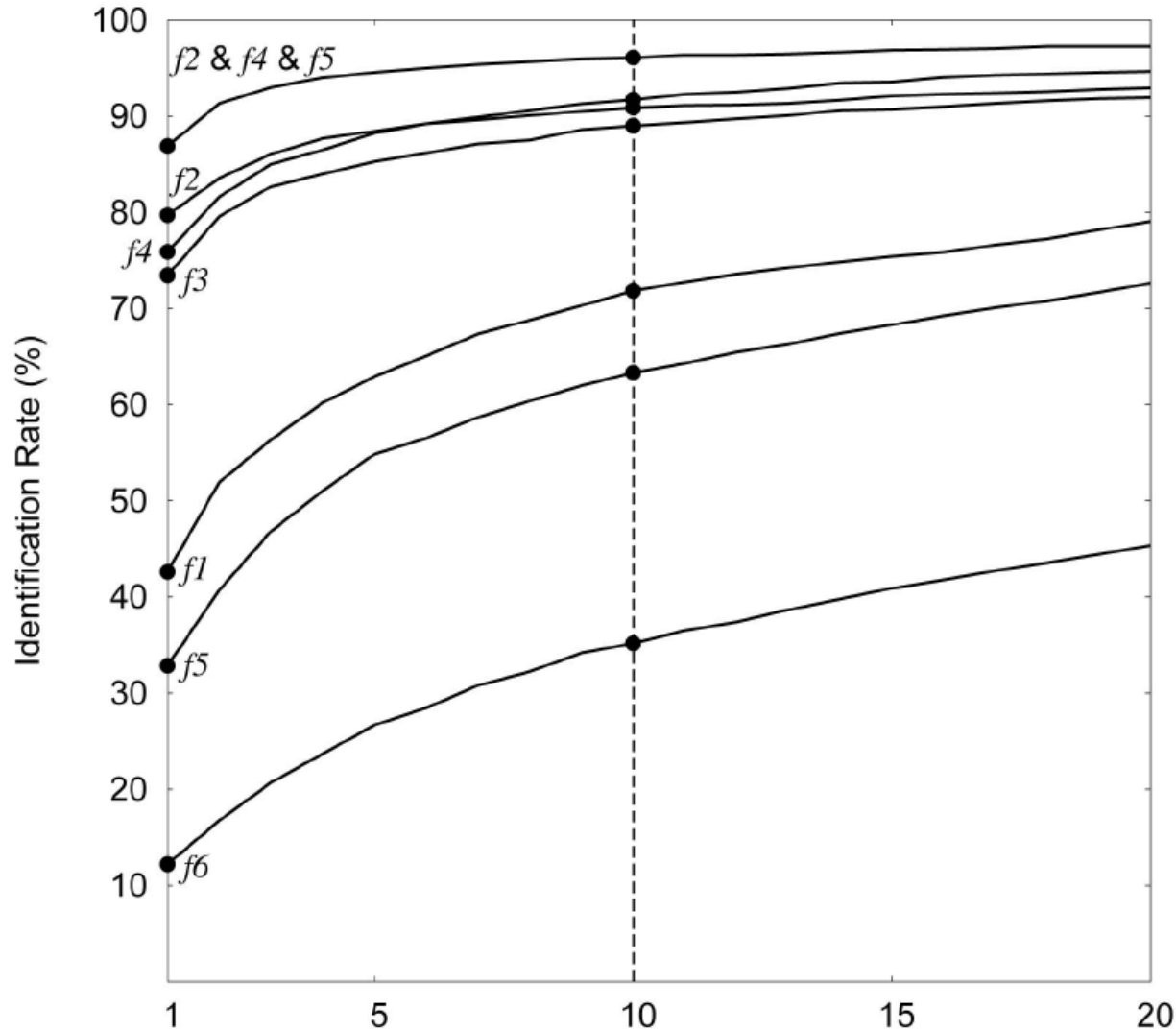
In the figure, for example:

- The correct identity is in the first position $\sim 78\%$ of the times
- We need an output list of 4 candidates to ensure that 95% of the times, the correct identity is in the list



Performance: Identification Mode (1:N)

Comparing different systems with CMC curves



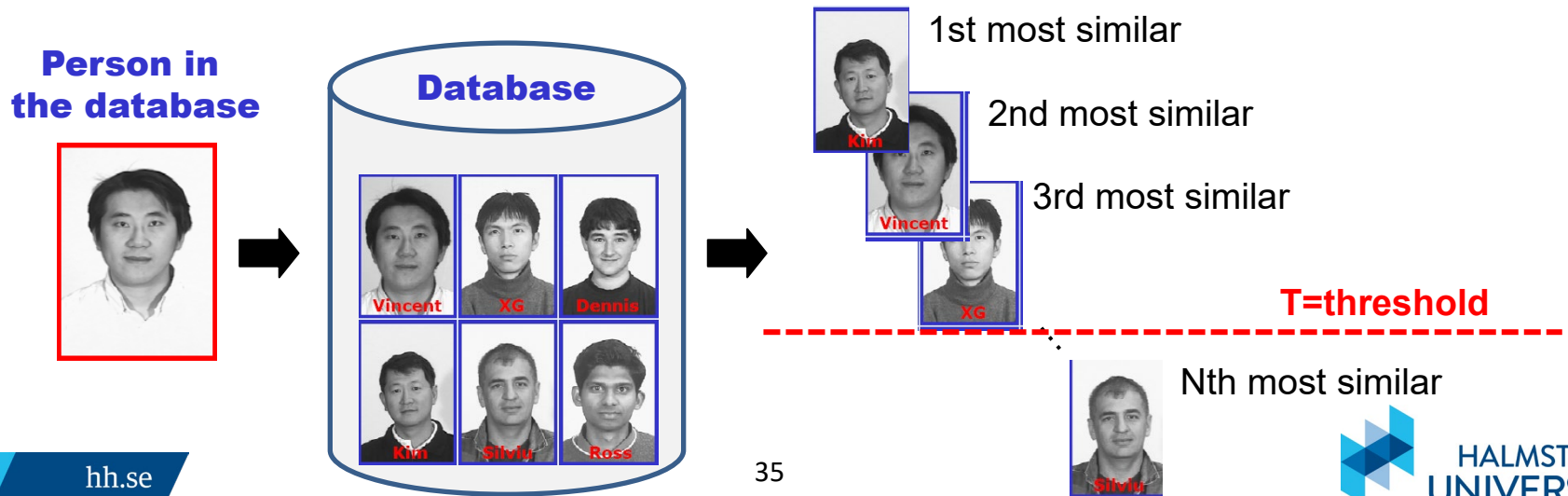
Performance: Identification Mode (1:N+1)

Identification (1:N+1 matching)

- The person trying to access is not necessarily in the database
- We add a threshold to discard the identities of the database which are not sufficiently similar to the input sample

If the person is in the database:

- **True Positive Identification Rate – TPIR(T)**: probability that the correct identity is within the **top M candidates** and the associated similarity score is above the threshold (or below if is a distance score)



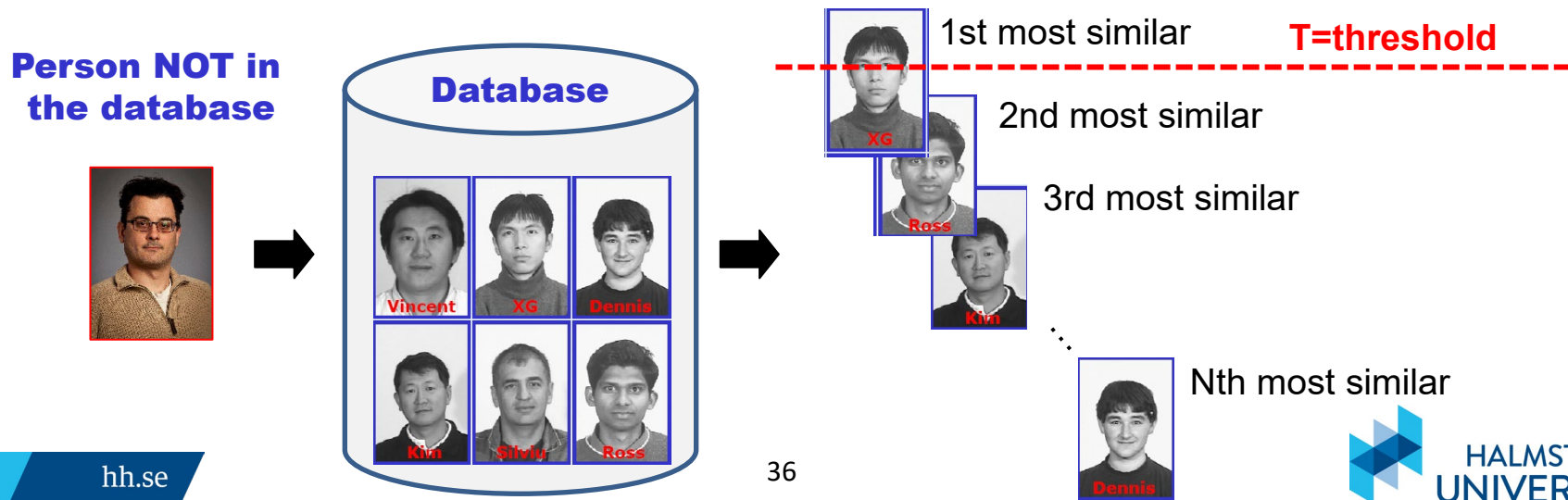
Performance: Identification Mode (1:N+1)

Identification (1:N+1 matching)

- The person trying to access is not necessarily in the database
- We add a threshold to discard the identities of the database which are not sufficiently similar to the input sample

If the person is NOT in the database:

- **False Positive Identification Rate – FPIR(T):** probability that there is a identity in the database whose associated similarity score is above the threshold (or below if it is a distance score)



Performance: Identification Mode

Impact of the database size

- 1:N (closed set): there is only 1 correct model and N-1 incorrect
- 1:N+1 (open set): all models of the database are incorrect

The error is expected to **increase with N** (size of the database)

- Person in the database = increase chances that other users in the database are ranked ahead in the list
- Person NOT in the database = increase chances that someone in the database surpass the threshold

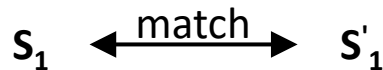
To avoid this problem when the database goes big, **indexing** strategies can be used, so the input signal is compared only with a portion of the database

Performance: connecting identification & verification

Simplistic approach to relate FAR/FRR in both modes

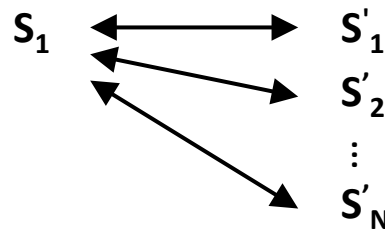
Verification (1:1)

FMR/FRR



Identification

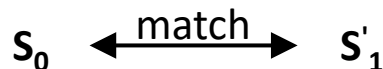
$FMR_N/FRR_N \approx FMR/FRR$



If the system is “good” in not rejecting genuine users (low FRR in verification mode), the right identity in identification mode will still come up high in the list when the user is in the database

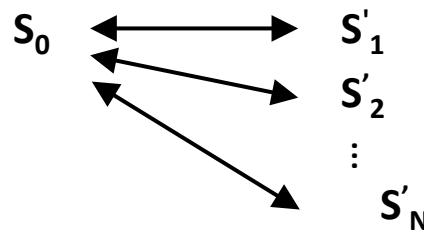
Verification (1:1)

FNMR/FAR



Identification

$FNMR_N/FAR_N \approx N \times FNMR/FAR$

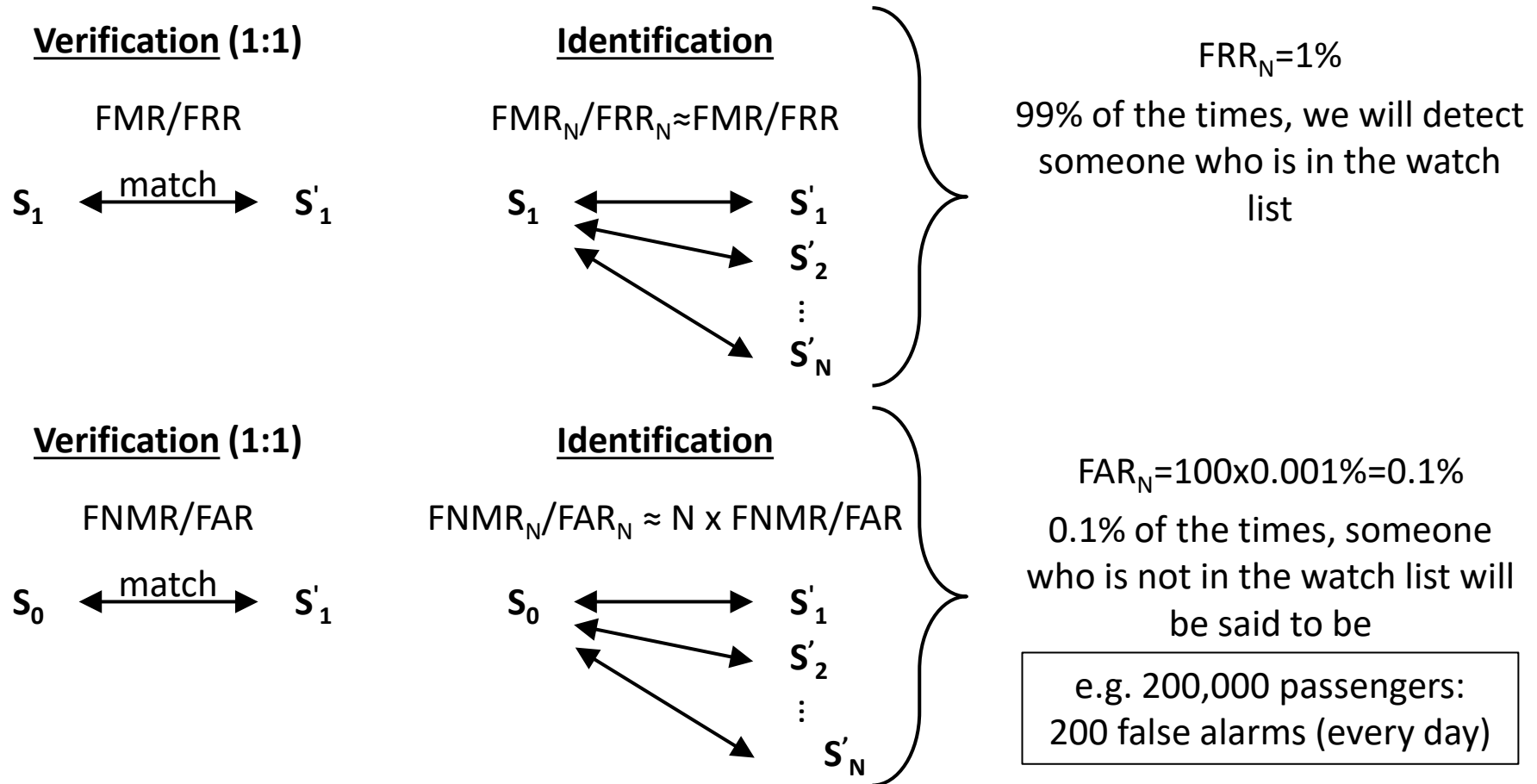


If the user is NOT in the database, the chances of finding a match in identification mode with someone in the database **increases proportionally with N**

Do not use identification if reducing false matches (FAR) is a priority, i.e. high security, because false matches increase with N

Performance: connecting identification & verification

Example: watch list with $N=100$, verification $FRR=1\%$, $FAR=0.001\%$



Do not use identification if reducing false matches (FAR) is a priority, i.e. high security, because false matches increase with N

Performance: Other Considerations

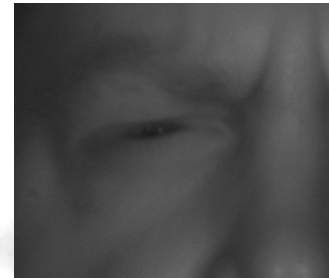
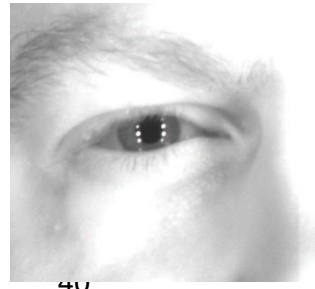
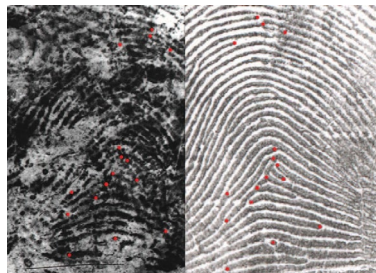
These errors are estimates based on a training database!

How the database is built has high impact on the estimates

If we use a different evaluation database, numbers will be different

If we use very "good" data in my evaluation database, we may get very good numbers, but the system will fail when "bad" data is presented

- Evaluation data should be representative of the conditions found during the operation of the system



Performance: Other Errors

Failure to Acquire (FTA)

- When the sensor fails to capture a biometric characteristic presented to it
- It can happen for example if a person does not have the characteristic (e.g. damaged finger) or if the sensor is not able to capture the sample (e.g. occluded face, finger removed too fast)

Failure to Enroll (FTE)

- When the biometric sample is captured, but it is not possible to generate a model (e.g. very low quality samples)
- FTE impacts on the perceived FAR/FRR: if we reject low quality samples, the perceived accuracy will increase

FTA and FTE are partly due to **intrinsic technological limitations** (e.g. the sensor is not good enough to capture certain samples)

Introduction to Biometric Technologies and Applications (II)

Kevin Hernández Díaz
ISLAB/CAISR

Email: kevin.hernandez-diaz@hh.se

Course: Biometric Recognition
DI4025, ht2025